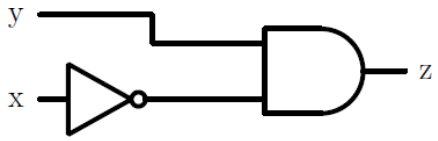
19.03.2018

EE 241 DIGITAL CIRCUITS
MIDTERM SOLUTION MANUAL

Name:

Q1) (18 points) A digital system has the following schematic form.

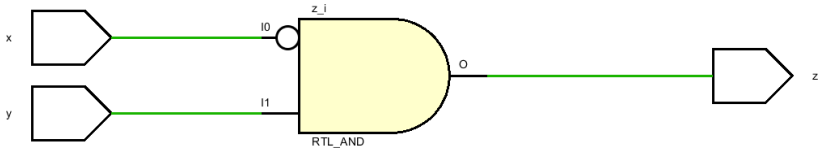


Represent this system in Verilog using

a- (9 points) Dataflow modeling.

Solution

```
module q1a (x,y,z);
input  x, y;
output z;
assign z=~x&y;
endmodule
```



b- (9 points) Structural modeling.

Solution

```
module q1b (x,y,z);
input  x,y;
output z;
wire w;
not gate_not(w,x);
and gate_and(z,w,y);
endmodule
```

Q2) (42 points) A digital system is represented as

```
module q2(x,y,z);
input [7:0] x;
output [7:0] y,z;

wire [7:0] yd,zd;

assign yd=x/10;
assign y=yd*10;

assign zd=x*10;
assign z=zd/10;

endmodule
```

What will be the value of output y and z (in decimal form), when the below testbench file is executed?

```
`timescale 1ns / 1ps

module q2tb;
reg [7:0] x;
wire [7:0] y,z;
q2 uut(x,y,z);
initial begin
x=8'b00011000; #10;
x=8'd96;  #10;
x=8'h7C;  #10;
$stop;
end
endmodule
```

Input	Output	
x	y	z
8'b00011000	20	24
8'd96	90	19
8'h7C	120	21

Q3) (40 points) Thermometer encoders find widespread use in Analog-to-Digital Converters (ADCs). It converts the raw data generated by the ADC into binary number. Table 1 shows the truth table of an 8-bit input 3-bit output thermometer encoder.

Table 1: Truth Table

i[7]	i[6]	i[5]	i[4]	i[3]	i[2]	i[1]	i[0]	o[3]	o[2]	o[1]	o[0]
0	0	0	0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0	0	1
1	1	0	0	0	0	0	0	0	0	1	0
1	1	1	0	0	0	0	0	0	0	1	1
1	1	1	1	0	0	0	0	0	1	0	0
1	1	1	1	1	0	0	0	0	1	0	1
1	1	1	1	1	1	0	0	0	1	1	0
1	1	1	1	1	1	1	0	0	1	1	1
1	1	1	1	1	1	1	1	1	0	0	0

a- (20 points) Write the Verilog description of the thermometer encoder in dataflow modeling with the truth table in Table 1.

b- (20 points) Specify a one-bit controller output (defined as "**output y;**" in Verilog) such that it will have logic level 1 when the converted value o[3:0] is greater than or equal to decimal value 4.

Solution

```
module q3(i,o,y);
input [0:7] i;
output [3:0] o;
output y;

//part a
assign o=i[7]+i[6]+i[5]+i[4]+i[3]+i[2]+i[1]+i[0];
//part b
assign y=i[4];
endmodule

`timescale 1ns / 1ps
module q3tb;
reg [0:7] i;
wire [3:0] o;
wire  y;
q3 uut(i,o,y);
initial begin
//i=7'b0011111; #10;
i=7'b0000111; #10;
$stop;
end
endmodule
```